Chromosome Research 1993, **1**, 107–115

Localization of seed protein genes on flow-sorted field bean chromosomes

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Received 15 March 1993;
Accepted for publication by Pat Heslop Harrison 10 April 1993.

Chromosomes from reconstructed field bean (*Vicia faba* L.) karyotypes were flow-sorted and the DNA was used for the physical localization of seed storage and nonstorage (USP) protein genes using PCR with sequence specific primers. The data were confirmed and refined by using DNA of microisolated chromosomes of other karyotypes as the target for PCR. The specificity of the PCR products was proved by restrictase digestion into fragments of predicted length or by reamplification using 'nested' primers. The genes are located within defined regions of chromosome I (USP = *u*nknown *s*eed *p*rotein genes), II (vicilin genes, legumin B3 genes), III (legumin B4 genes), IV (pseudogenes Ψ1) and V (legumin A genes and pseudogenes Ψ1). Except for the pseudogene derived from the sequence of legumin B4 gene, all members of each gene family are located in one chromosome region exclusively. This approach proved to be useful for localizing genes that cannot be mapped genetically (due to the lack of allelic variants) and might be applied to integrate physical and genetic maps.

Key words: *Vicia faba*, chromosome flow sorting, gene mapping, PCR, seed protein genes

Introduction

Physical gene mapping is important for determination of physical distances within genetic maps and essential for localization of genes which cannot be localized by genetic methods due to the lack of true allelic variants, as is the case for most seed storage protein genes. Knowledge of physical gene location is helpful for genetic manipulations in basic and breeding research. Physical mapping may be performed by hybridizing labelled probes *in situ* (if the targets are not too small to detect) or to the DNA of chromosomes isolated in bulk (Collard *et al.* 1985). An alternative is PCR amplification of the gene in question using sequence specific primers and DNA of defined microisolated (Macas *et al.* 1993) or flow sorted chromosomes (Cotter *et al.* 1989). The former method is more laborious; the latter was up to now not applicable for plants since it was difficult to obtain sufficiently pure suspensions of metaphase chromosomes distinctive enough for successful sorting of all chromosomes of a plant genome (Arumuganathan *et al.* 1991, Wang *et al.* 1992). Here we describe a method which allows physical mapping of single or low copy sequences on flow sorted field bean chromosomes by PCR, and define the chromosome regions of hitherto unlocalized seed protein genes.

Material and methods

Isolation of genomic DNA

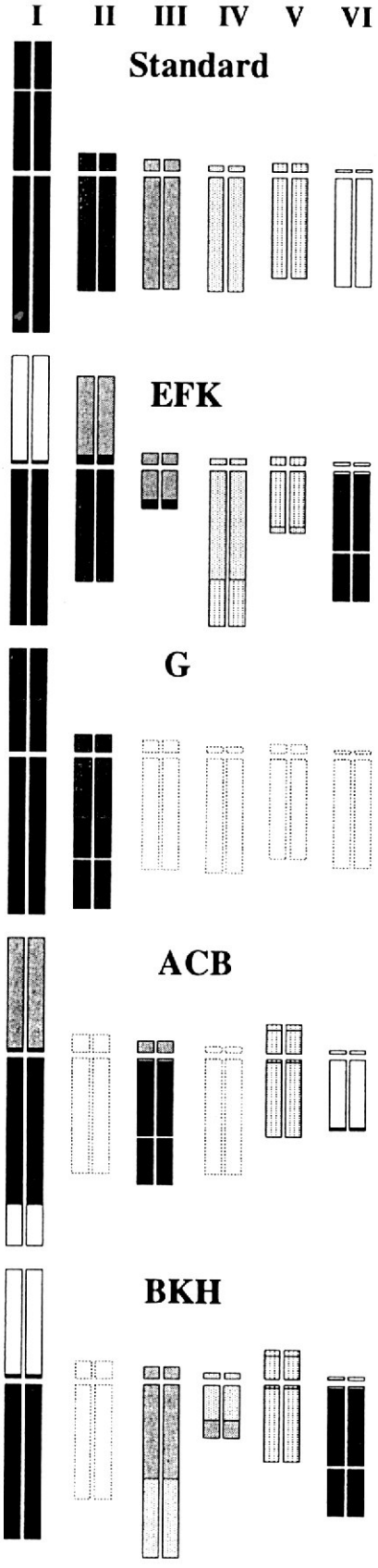
Genomic DNA was isolated from *V. faba* leaves as described (Dellaporta *et al.* 1983). Because of the high content of polyphenolics in field bean plants the extraction buffer additionally contained 1% polyvinylpyrrolidone 25 (Serva, Heidelberg). A second precipitation step was performed after additional treatment with phenol/chloroform/isoamylalcohol (25 : 24 : 1) using 1 vol isopropanol.

Isolation of chromosomes

Suspensions of metaphase chromosomes were obtained from the wildtype and from the reconstructed karyotypes EF, EFK and G of *V. faba* ($n = 6$) after synchronization of root tip meristems by hydroxyurea

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and colchicine treatment as described (Dolezel *et al.* 1992, Schubert *et al.* 1993). By virtue of their different sizes, these karyotypes can be sorted by fluorescence activated cell sorting (FACStar Plus, Becton Dickinson, Heidelberg) into chromosome I (wildtype); I, II, III, IV, V (EFK); I, II, III, IV, V, VI (EF) and I, II (G). For description of the karyotypes see Schubert & Rieger (1991) and Figure 1. For chromosome fractionation by flow cytometry on the basis of different DNA content, chromosome suspensions were stained with propidium iodide and handled as described by Lucretti *et al.* (1993). Micromanipulation of chromosomes from squash preparations of karyotypes ACB and BKH (Figure 1) was performed according to Schubert *et al.* (1993).

Conditions of the polymerase chain reaction

PCR was performed using microisolated chromosomes (Macas *et al.* 1993). The procedure was slightly modified for flow sorted chromosomes: 200 or 300 chromosomes were sorted into 59 µl of sterile, deionized water in PCR reaction tubes and stored at –20°C. Before use, 40 µl of 2.5 × PCR mix was added. The final concentration of the reagents was: 0.2 mM dNTP, 0.2 µM primers, 2.0 mM $MgCl_2$, 50 mM KCl, 10 mM Tris-HCl (pH 9.0 at 25°C), 0.1% Triton X-100. Samples were overlaid with mineral oil and denatured at 94°C for 10 min. After denaturation the temperature was decreased to 90°C, 1 µl (5 U) of *Taq* DNA polymerase (Promega, Heidelberg) was added and 35 or 40 cycles of amplification (55°C 0.8 min, 72°C 1 min, 94°C 0.8 min) with the final primer extension step for 5 min were performed in a Perkin Elmer DNA thermal cycler 480. Restriction enzymes (Boehringer, Mannheim) were applied according to supplier's instructions. The amplified products or their restrictase digestion fragments were electrophoresed on 1% agarose gels. The designation of sequence-specific

Figure 1. Scheme of the standard (wildtype) and of reconstructed karyotypes of *V. faba* used for flow sorting or microisolation of certain chromosome types. Various symbols within reconstructed chromosomes reflect their composition of parts of wildtype chromosomes. Karyotype EF is omitted since except for translocation K, substituting wildtype chromosomes I and VI by their translocation products in karyotype EFK, EF is identical to EFK. The following chromosomes were isolated by flow sorting: Chromosome I of karyotypes standard/EF, EFK, and G; chromosome II of karyotypes EF/EFK and G; chromosomes III, IV, and V of karyotypes EF/EFK and chromosome VI (wildtype) of karyotype EF. The following chromosomes were separately microisolated: Chromosome I of karyotype ACB; chromosomes II, III, IV, and V of karyotype BKH. The remaining chromosomes of these karyotypes, 20 each, were pooled together for control PCR.

Table 1. Primers used for studies of seed protein gene location

Gene	Sequence	Positions	Remarks
vicilin (vic1)[1]	5′ GGC,AGC,TAC,CAC,ATT,GAA,AGA,TTC,ATT,TTC 3′	42–71	841 bp including
	5′ CCA,GAT,AAG,ATG,GAC,GGT,TGG,TTT,TGA,TTA,CC 3′	882–851	two introns
USP[2]	5′ GAA,GAT,TAT,TGG,CAA,TCT,ATT,TGG,CC 3′	208–233	826 bp incl.
	5′ TTG,TCT,GCG,GTA,TGA,TTT,GGT,ACC,C 3	1033–1009	2nd intron
leg A[3]	5′ CCA,ATG,CTC,CCC,AAG,AAA,TTT,ACA,TCC 3	277–303	694 bp + 2 introns
(cDNA)	5′ CCA,TTG,TCT,CCA,TGC,CTT,CTG,CTT,TCA,CC 3′	970–942	
nested primer	5′ GGC,TTC,AAG,AGG,GAT,TTC,TTG,GAA,GAT,GC 3′	663–691	308 bp
leg B4[4]	5′ TTA,CAT,CAT,CCA,AGG,AAA,GGG,TGT,TAT,TGG 3′	335–364	1029 bp incl. 2 introns
	5′ GCG,TTT,ATG,TTC,CAG,TGT,GGA,GCA,TAT,ATA,CC 3	1363–1332	or 332 bp (Ψ1)
nested primer	5′ CCT,TCC,CTG,CTC,ACT,TCT,CTG,CTT,CTC,CTC 3′	1057–1028	723 bp incl. 1 intron
Leg B3[5]	5′ TTC,ATC,ATT,CAA,GGG,AAG,GGA,GTA,CTT,GG 3′	118–146	1172 bp +
(minor)	5′ GCG,TTG,ATG,TTC,CAG,TGT,GGA,GCA,TAT,ATA,CC 3′	1289–1258	unknown intron
nested primer	5′ CCG,AAA,CAC,AGG,AAC,AAC,ATC,AAC,AAA,GGC 3	416–445	875 bp

References: [1] Weschke *et al.* (1988)
[2] Bäumlein *et al.* (1991)
[3] Schlesier *et al.* (1990)
[4] Heim *et al.* (1989)
[5] Heim and Wobus (unpublished)

primers is shown in Table 1 and Figure 2.

Results

Subchromosomal localization of genes coding for USP, legumin A, legumin B3, B4 and of a pseudogene of the legumin B4 gene group on flow sorted chromosomes

Nine types of chromosomes were obtained from various karyotypes by direct sorting into PCR tubes (wildtype I, VI; EF II, III, IV, V; EFK I; G I & II). Their DNA was used as a target for PCR (35 cycles of amplification each) with primers designed according to the genomic or cDNA sequences of the corresponding genes. The expected size of amplified fragments in base pairs was: 826 for the USP gene, 1029 for the legumin B4 gene, 332 for the corresponding pseudogene, 1172 plus unknown intron 1 for the legumin B3 gene, 694 plus two introns of unknown length for the legumin A gene and 841 for the vicilin gene which served as a control (Table 2).

Using vicilin gene-specific primers, a band of the expected size was obtained with DNA from all tested chromosome types. However, only DNA from chromosomes II of karyotypes G and EFK, which previously (Macas *et al.* 1993) were shown to harbour the only locus amplifiable by the same primers, yielded a strong band (data not shown). USP gene-specific primers resulted in a very strong band of 826 bp when the DNA of chromosome I of wildtype or of karyotype G was used as a target (Figure 3a).

Two adjacent strong bands of about 1000 bp were obtained only with legumin B4 gene-specific primers and DNA of chromosome III of karyotype EFK, indicating two (groups of) genes located on the same chromosome. The same primers yielded a strong band of 332 bp corresponding to the pseudogene of the legumin B4 gene group (*leB* Ψ1, Heim *et al.* 1989) only after reaction with chromosome IV of karyotype EFK (Figure 3b).

Primers designed according to the cDNA of legumin A led to amplification of three strong bands within the range of 800 to 1000 bp exclusively with DNA from chromosome IV of karyotype EFK (Figure 3c). Primers designed according to the sequence of legumin B3 gene resulted in two narrow bands of about 1.3 kb with DNA from chromosome III of karyotype EF (Figure 3d) and from chromosome II of karyotype G (data not shown).

These results indicate that, due to the high efficiency of PCR (only a few chromosomes are needed to yield positive signals with gene-specific primers, Macas *et al.* 1993), contamination of a chromosome fraction by a few chromosomes harbouring the gene in question may result in a weak amplification fragment. Microscopic controls of 50-200 chromosomes of each type sorted on cover slips revealed contamination by mis-sorted chromosomes, broken arms or satellites of between 0 and 10% (average 6%). In spite of that, the distribution and strength of bands allowed the following prediction of gene loci. The USP genes should be located on the long arm of chromosome I. The genes coding for legumin B4 should be on the proximal part of chromosome III, since the distal half of the long arm of chromosome III of karyotype EFK is substituted by half of the short arm of chromosome II, but chromosome II of karyotype G, where the short arm is in original position, did not result in amplification of a band with legumin B4 primers. Genes coding for legumin B3 should be located on the

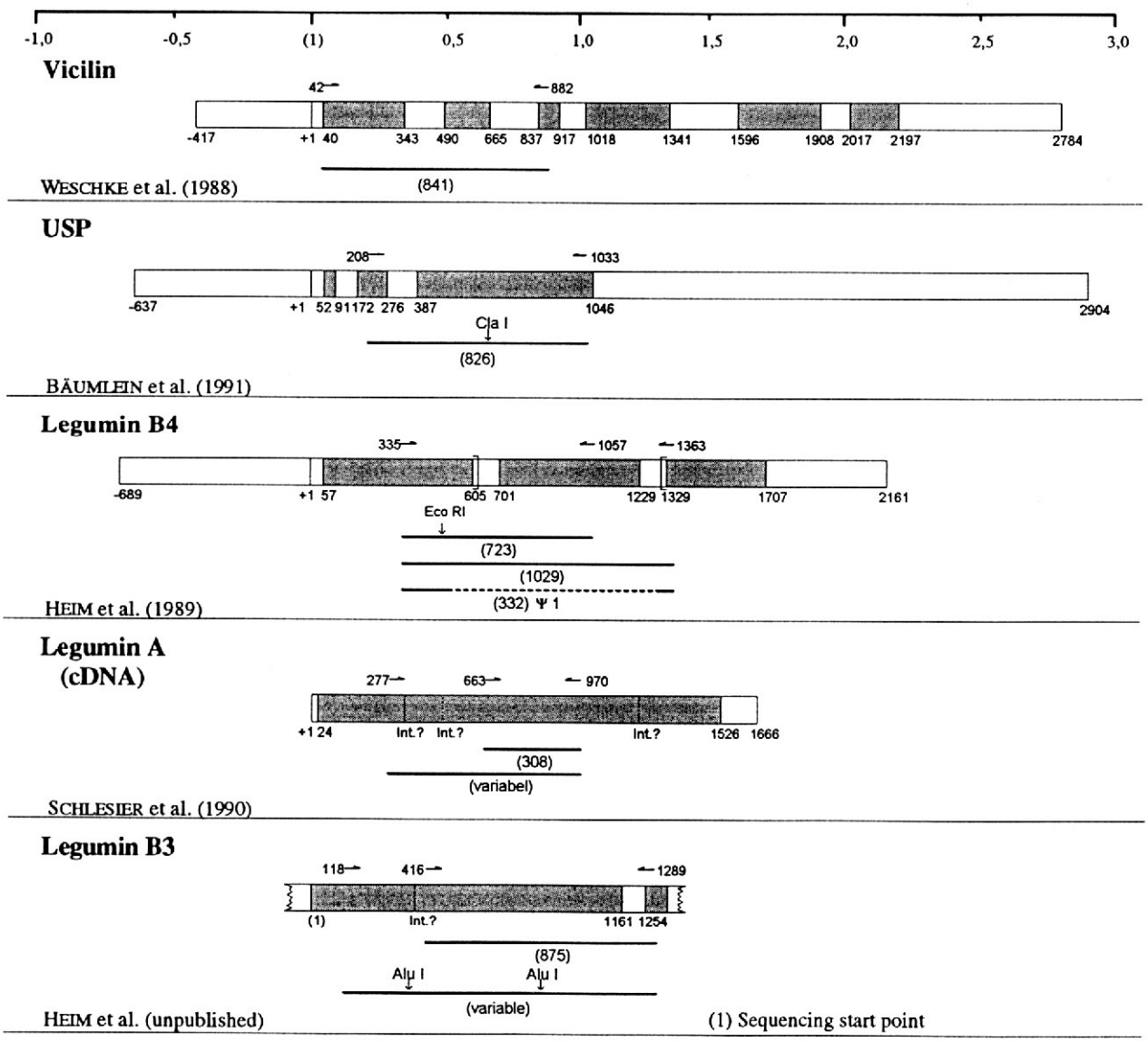


Figure 2. *V. faba* seed storage protein genes shown schematically, the designation of primers (horizontal arrows) the expected amplification products (thick lines) and selected restrictase cleavage sites therein (vertical arrows).

distal half of the short arm of chromosome II since chromosome III of karyotype EF as well as chromosome II of karyotype G revealed a typical band. The pseudogene of legumin B4 seemed to be located either on chromosome IV or on the distal part of the long arm of chromosome V, which is translocated to chromosome IV of karyotype EFK. The same is true for the genes coding for legumin A.

Proof and further specification of the predicted gene loci by PCR on microisolated chromosomes

Suitable chromosomes were isolated by micromanipulation from different karyotypes and tested by PCR. All other chromosomes of the corresponding karyotypes (20 of each type) pooled together served as a control. The results are summarized in Table 3 and Figures 4 and 5. Twenty chromosomes I of karyotype ACB revealed a 826 bp band when submitted to PCR with USP gene-specific primers, while the remaining chromosomes of that karyotype did not (Figure 4). Genes coding for USP are therefore located within a region of the long arm of chromosome I, including the centromere but excluding its very distal part which in karyotype ACB is substituted by one-third of the long arm of wildtype chromosome VI.

Legumin A gene-specific primers yielded the typical bands exclusively when DNA of the submetacentric chromosome V of the karyotype BKH was the target (Figure 4). Thus, the distal third of the long arm of chromosome V, which is translocated to chro-

Table 2. Results of PCR reactions (35 cycles of amplification each) with the DNA of flow-sorted chromosomes as a target and primers designed according to genomic or cDNA squences of genes coding for vicilin, USP, legumin B3, B4, legumin A, and of pseudogene Ψ1 derived from legumin B4 gene

Chromosome	Karyotype	Number of chromosomes	Vicilin	USP	Legumin B3	B4	Ψ1	Legumin A
I	wildtype = EF	300	(+)	++	–	(+)	(+)	(+)
I	EFK	200	(+)	+	nt	–	–	(+)
II	EFK = EF	200	++	nt	–	–	(+)	(+)
III	EFK = EF	300	nt	nt	++	+–	–	(+)
IV	EFK = EF	200	(+)	+	–	–	+–	++
V	EFK = EF	300	(+)	+	–	–	(+)	(+)
I	G	200	(+)	++	(+)	(+)	(+)	–
II	G	200	++	+	++	–	–	–
VI	wildtype = EF	200	nt	nt	–	nt	nt	nt

nt, not tested; –, no signal; (+), very weak band; +, weak band; ++, strong band of the expected size

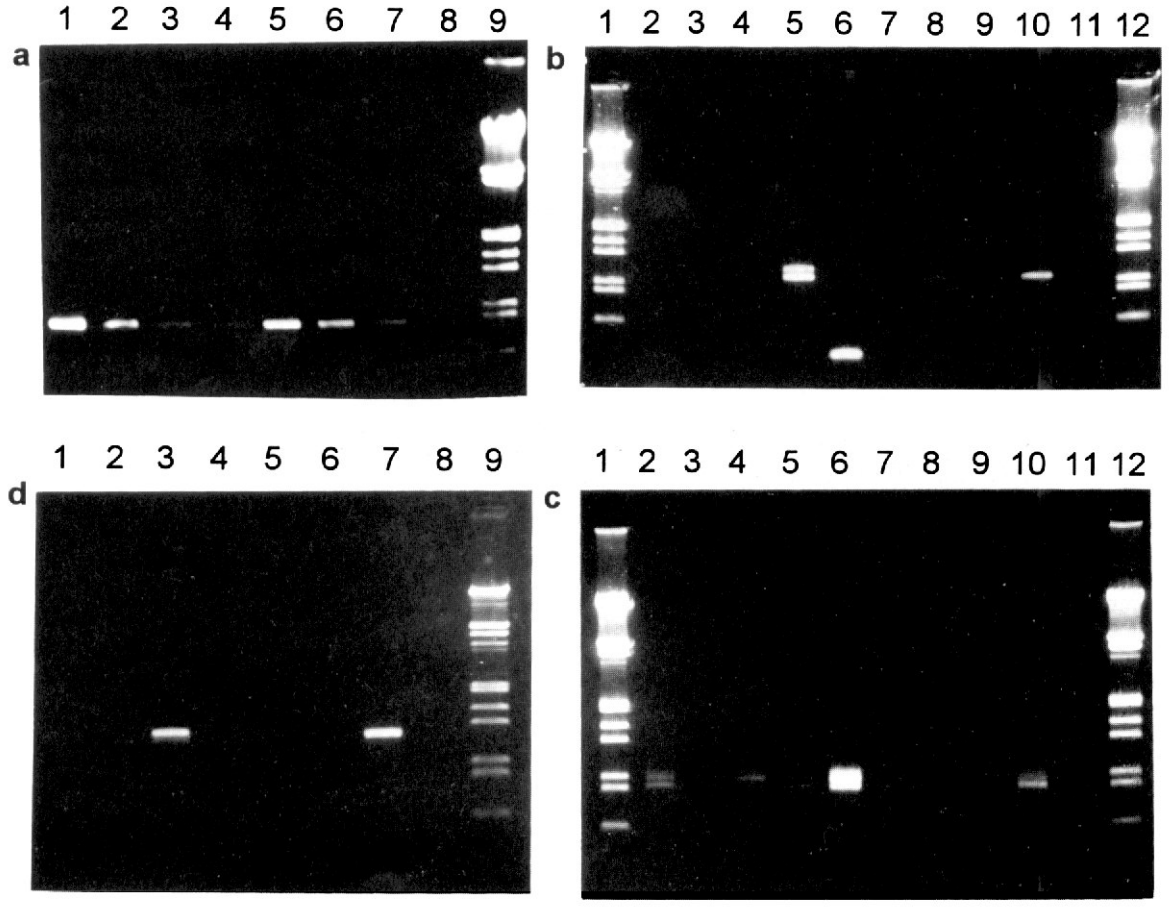


Figure 3. Agarose gel electrophoresis of PCR products obtained with gene-specific primers and DNA of flow sorted field bean chromosomes. a) USP gene-specific primers: (1) wildtype chromosome I, (2-4) chromosomes I, IV and V of karyotype EFK (5,6) chromosomes I and II of karyotype G, (7) 125 pg of total genomic DNA of *V. faba*, (8) no DNA, (9) size markers (*EcoR*I/*Hind*III digested λ DNA). b) Legumin B4 gene-specific primers (1,12) size markers, (2) wildtype chromosome I, (3-7) chromosomes I, II, III, IV, V of karyotype EFK, (8,9) chromosomes I and II of karyotype G, (10) genomic DNA, (11) no DNA. (The band below the smallest marker band corresponds to the pseudogene Ψ1). c) Legumin A gene-specific primers. The sequence of slots as in b). d) Legumin B3 gene-specific primers. (1-6) chromosomes I, II, III, IV, V and VI of karyotype EF, (7) genomic DNA, (8) no DNA, (9) size markers.

Table 3. Results of PCR reactions (40 cycles of amplification) made with the DNA of microisolated chromosomes (20 of each) from various karyotypes and primers specific for the genes of the USP, legumin B3, B4/Ψ1 and legumin A

Chromosome	Karyottype	Occurrence of the amplified fragments				
		USP	Legumin B3	B4	Ψ1	A
I	ACB	+				
II III IV V VI	ACB	–				
V	BKH					+
I II III IV VI	BKH					–
III	BKH			+	–	
IV				–	+	
V				–	(+)	
I II VI	BKH			–	–	
II	BKH		+			
III	BKH		–			
I IV V VI	BKH		–			

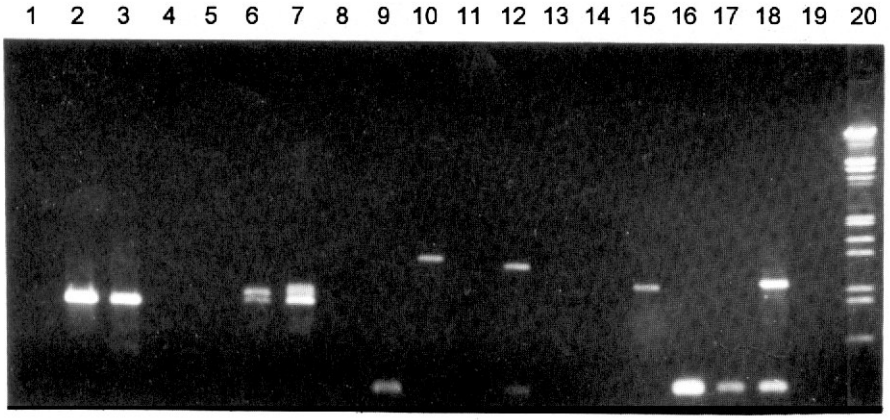


Figure 4. Agarose gel electrophoresis of PCR products obtained with gene-specific primers and DNA of microisolated chromosomes. USP gene-specific primers: 1) all chromosomes (20 each), of karyotype ACB except I, 2) 20 chromosomes I of the same karyotype, 3) genomic DNA, 4) no DNA. Legumin A gene-specific primers: 5) all chromosomes (20 each), except V, of karyotype BKH, 6) 20 chromosomes V of the same karyotype, 7) genomic DNA, 8) no DNA. Legumin B3 gene-specific primers: 9) all chromosomes (20 each), except II and III, of karyotype BKH, 10) 20 chromosomes II, 11) 20 chromosomes III of karyotype BKH, 12) genomic DNA, 13) no DNA. (The small band in 9) and 12) indicates a pseudogene derived from the sequence of legumin B3 genes.) Legumin B4 gene-specific primers: 14) chromosomes I, II and VI (20 each) of karyotype BKH, 15) 20 chromosomes III, 16) 20 chromosomes IV, 17) 20 chromosomes V of the same karyotype, 18) genomic DNA, 19) no DNA, 20) size markers as in Figure 3. (The small fragment in slots 16) to 18) represents the pseudogene Ψ1 derived from legumin B4 genes.)

mosome IV of karyotype EFK, contains the genes coding for legumin A.

The location of the legumin B4 genes within the proximal part of chromosome III was confirmed using DNA of chromosome III of karyotype BKH as a target (Figure 4). The 332 bp band representing the pseudogene Ψ1 appeared when DNA of chromosome IV or V (weaker band) of karyotype BKH was used (Figure 4). Thus, most members of this pseudogene family are located at the proximal part of chromosome IV and a minority at the distal part of the long arm of chromosome V, where genes coding for legumin A are also located.

Legumin B3 gene-specific primers yielded a corresponding band only after PCR with DNA of chromosome II of karyotype BKH (Figure 4). Therefore, these

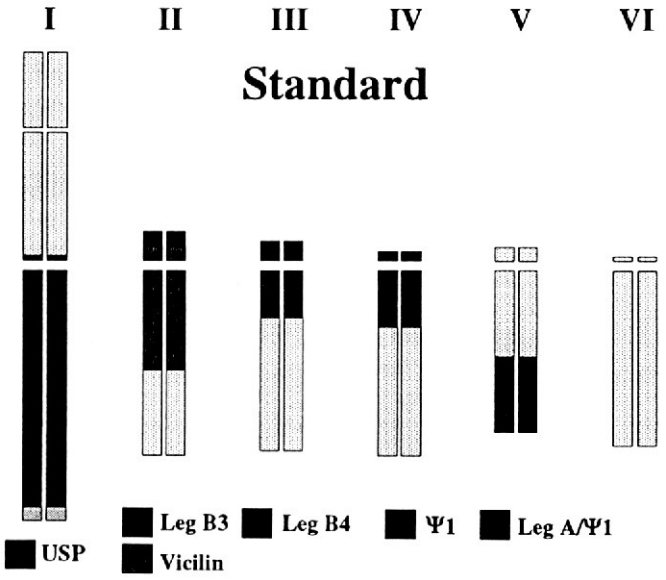


Figure 5. Scheme of the standard (wildtype) karyotype of the field bean with those regions marked to which the seed protein genes (or their pseudogenes) were assigned.

genes are located at the distal part of the short arm of chromosome II, which is translocated to chromosome III in karyotype EF.

Evidence for the specificity of the amplified fragments by restrictase digestion or 'nested' PCR

The fragment resulting from chromosome I of karyotype ACB after PCR with USP gene-specific primers was nearly completely digested by *Cla*I into two fragments of 381 and 446 bp, respectively (Figure 6a), as expected from the position of the recognition site for this restriction enzyme within the inter-primer sequence.

The PCR fragments corresponding to the genes coding for legumin A from chromosome V of karyotype BKH were reamplified using the 3′-primer of the first experiment and a second 5′-primer according to a sequence of the first inter-primer region behind both introns (nested primer) and resulted, as expected, in a single fragment of 308 bp (Figure 6b).

The fragments obtained from PCR with legumin B3

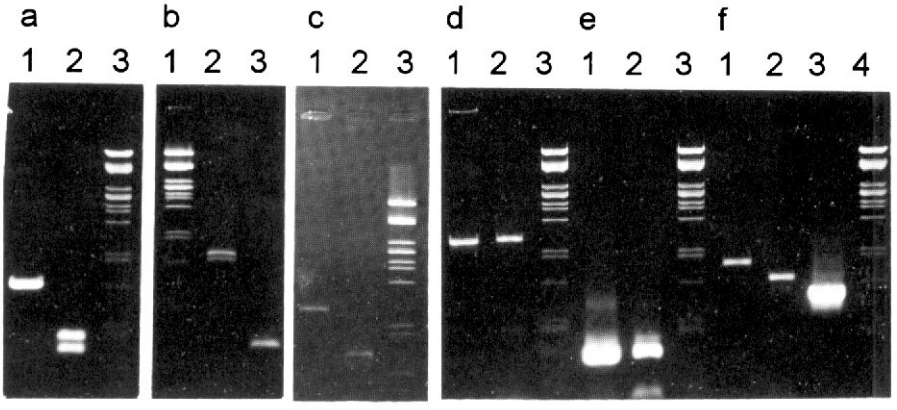


Figure 6. Agarose gel (1.5%) electrophoresis of restrictase cleavage fragments or reamplification products with nested primers of PCR products obtained with gene-specific primers and DNA of defined chromosomes. a) USP: 1) PCR product obtained with DNA of flow sorted wildtype chromosomes I (Figure 3a, slot 1); 2) the same after *Cla*I digestion; 3) size markers (*Pst* I-digested λDNA). b) Legumin A: 1) size markers; 2) PCR products obtained with 20 chromosomes V of karyotype BKH (Figure 4, slot 6); 3) the same after reamplification with a nested 5′-primer. c) Legumin B3: 1) PCR products obtained with DNA of flow sorted chromosomes III of karyotype EF (Figure 3d, slot 3); 2) the same after reamplification with a nested 5′-primer; 3) size markers. d) Legumin B3: 1) PCR products as in c); 2) the same after digestion with *Alu*I; 3) size markers. e) Pseudogene Ψ1: 1) PCR products obtained with 20 chromosomes IV of karyotype BKH (Figure 4, slot 16); 2) the same after digestion with *EcoR*I; 3) size markers. f) Legumin B4: 1) PCR product obtained with 20 chromosomes III of karyotype BKH (Figure 4, slot 15); 2) the same after digestion with *EcoR*I; 3) the same after reamplification with a nested 3′-primer; 4) size markers.

gene-specific primers and DNA of chromosome III of karyotype EF were reamplified using a nested 5'-primer and resulted in 875 bp fragment, as expected, and in a weak smaller fragment (Figure 6c). The same PCR product was also partially digested by *Alu*I and yielded three fragments (240 bp in front of intron 1, 439 bp including intron 2 and 492 bp plus intron 1) as expected according to the two inter-primer recognition sites (Figure 6d).

The fragments obtained with the legumin B4 gene-specific primers and DNA of chromosomes III and IV of karyotype BKH, respectively, were after reamplification digested by *Eco*RI. Due to a single recognition site within the inter-primer sequence the pseudogene fragment (chromosome IV) is at least partially cleaved into 170 and 161 bp which appear as one band (Figure 6e), while the fragments corresponding to the coding genes are decreased in size by 170 bp (Figure 6f). Additionally, reamplification of the legumin B4 gene specific fragments using a nested 3'-primer designed according to the sequence between both introns resulted in a 723 bp fragment as expected (Figure 6f).

Discussion

For the first time flow-sorted plant chromosomes have been successfully used for physical localization of genes using PCR and sequence-specific primers. Since flow-sorted chromosome fractions are rarely 100% pure (in these experiments the average purity was 94%), they usually contain some mis-sorted chromosomes. This contamination may result in the occurrence of weak PCR bands from chromosome fractions which do not harbour the gene in question. This is critical when low copy number sequences (such as seed storage protein genes) have to be localized. In such cases the location of single members of the corresponding gene family at other chromosomal positions has to be excluded. Therefore, experiments for confirmation and refinement of localization were performed on microisolated chromosomes from other karyotypes, since microisolation ensures absolute purity and 20 chromosomes are sufficient to detect a single gene copy (Macas *et al.* 1993). In each case the predictions derived from the first set of experiments with flow sorted chromosomes were confirmed.

The genes coding for USP, legumin A, B3, B4 and of the pseudogene Ψ1 were finally localized within defined chromosome regions. The previously reported location of genes coding for legumin A on chromosome I and for legumin B4 on chromosome V based on *in situ* hybridization with radioactively labelled cDNA probes (De Pace *et al.* 1991) was not confirmed and the PCR technique proved to be more reliable than *in situ* hybridization for mapping of sequences < 1 kb in length. The sequence specificity of the PCR products could be proved by digestion with restrictases into fragments of predicted length or by reamplification using nested primers.

The observed degree of variability in number and/or size of the PCR products (e.g., for legumin A, B3 and B4 genes between karyotypes EF, EFK and BKH) reflected the natural polymorphism characteristic of the members of this gene family among individuals within and between different lines of the field bean (Pich & Schubert in press). This is paralleled by sequence, Southern and protein data (Heim *et al.* 1989, Schlesier *et al.* 1990, Horstmann *et al.* 1993).

The experimental system developed in this study is applicable to all plant species with chromosomes distinct enough for flow sorting, by nature or after reconstruction of their karyotypes (e.g. barley, Künzel & Nicoloff 1979), provided pure chromosome suspensions are obtainable. It is expected that in future, signals derived from chromosome-specific nucleic acids or proteins will allow the discrimination and sorting of single chromosome types, even from karyotypes with morphologically similar chromosomes.

Further decrease in number of chromosomes per sorted fraction (~50) will reduce the absolute number of mis-sorted chromosomes without diminishing sensitivity. According to our observation, mis-sorting of chromosomes does not occur at random. It usually involves chromosomes of similar size, broken arms or satellites similar in DNA content to small chromosomes, or small chromosomes that have adhered to each other and have a similar DNA content to larger chromosomes. Therefore, intentional selection of chromosomes from two or three differently reconstructed karyotypes provides valid controls and will certainly avoid misinterpretation when single copy sequences have to be assigned to flow sorted chromosome fractions.

It is concluded that about 10 fractions of flow-sorted chromosomes from three differently reconstructed karyotypes with chromosomes distinct enough for sorting allow the localization of any unique sequence at a subchromosomal level. This will permit the rapid construction of a physical map of *V. faba* and the integration of preliminary linkage maps (Van De Ven *et al.* 1990, 1991) which up to now have not been chromosomally assigned.

Acknowledgements

We are grateful to Ute Heim and Ulrich Wobus for providing unpublished sequence data of legumin B3 genes, to Bernhard Schlesier and Rosemarie Gillandt for synthesis of primers, to Barbara Hildebrandt, Joachim Bruder and Jiřína Eliášova for technical assistance and to Rigomar Rieger for critical reading of the manuscript. This work was supported in part by the Deutsche Forschungsgemeinschaft (Schl-321/1-1).

References

Arumuganathan K, Slattery JP, Tanksley SD, Earle ED (1991) Preparation and flow cytometric analysis of metaphase chromosomes of tomato. *Theor Appl Genet* **82**: 101-111.

Bäumlein H, Boerjan W, Nagy I, *et al.* (1991) A novel seed protein gene from *Vicia faba* is developmentally regulated in transgenic tobacco and *Arabidopsis* plants. *Mol Gen Genet* **225**: 459-467.

Collard JG, de Boer PAJ, Janssen JWG, Schijven JF, de Jong B (1985) Gene mapping by chromosome spot hybridization. *Cytometry* **6**: 179-185.

Cotter F, Nasipuri S, Lam G, Young BD (1989) Gene mapping by enzymatic amplification from flow-sorted chromosomes. *Genomics* **5**: 470-474.

Dellaporta SL, Wood J, Hicks JB (1983) A plant molecular DNA minipreparation. Version II. *Plant Mol Biol Rep* **1**: 19-21.

De Pace C, Delre V, Scarascia Mugnozza GT, *et al.* (1991) Legumin of *Vicia faba* major: Accumulation in developing cotyledons, purification, mRNA characterization and chromosomal location of coding genes. *Theor Appl Genet* **83**: 17-23.

Doležel J, Číhalíková J, Lucretti S (1992) A high yield procedure for isolation of metaphase chromosomes from root tips of *Vicia faba* L. *Planta* **188**: 93-98.

Heim U, Schubert R, Bäumlein H, Wobus U (1989) The legumin gene family: structure and evolutionary implications of *Vicia faba* B-type genes and pseudogenes. *Plant Mol Biol* **13**: 653-663.

Horstmann C, Schlesier B, Otto A, Kostka S, Müntz K (1993) Polymorphism of legumin subunits from field bean (*Vicia faba* L. var. minor) and its relation to the corresponding multigene family. *Theor Appl Genet* **86**: in press.

Künzel G, Nicoloff H (1979) Further results on karyotype reconstruction in barley. *Biol Zentbl* **98**: 587-592.

Lucretti S, Doležel J, Schubert I, Fuchs J (1993) Flow karyotyping and sorting of *Vicia faba* chromosomes. *Theor Appl Genet* **85**: 665-672.

Macas J, Weschke W, Bäumlein H, *et al.* (1993) Localization of vicilin genes via polymerase chain reaction in microisolated field bean chromosomes. *Plant J*, in press.

Pich U, Schubert I (1993) Polymorphism of legumin genes in inbred lines of *Vicia faba*. *Biol Zentbl* **112**: in press.

Schlesier B, Bassüner R, Hai NV, Müntz K (1990) The cDNA derived primary structure of two distinct legumin A subunit precursors from field bean (*Vicia faba* L.). *Nucleic Acids Res* **18**: 7146.

Schubert I, Rieger R (1991) Catalogue of chromosomal and morphological mutants of faba bean in the Gatersleben Collection, 1991. *FABIS Newslett* **28/29**: 14-22.

Schubert I, Doležel J, Houben A, Scherthan H, Wanner G (1993) Refined examination of plant metaphase chromosome structure at different levels made feasible by new isolation methods. *Chromosoma* **102**: 96-101.

van de Ven M, Powell W, Ramsay G, Waugh R (1990) Restriction fragment length polymorphisms as genetic markers in *Vicia*. *Heredity* **65**: 329-342.

van de Ven WTG, Waugh R, Duncan N, Ramsay G, Dow N, Powell W (1991) Development of a genetic linkage map in *Vicia faba* using molecular and biochemical techniques. *Aspects Appl Biol* **27**: 49-54.

Wang ML, Leitch AR, Schwarzacher T, Heslop-Harrison JS, Moore G (1992) Construction of a chromosome-enriched *Hpa*II library from flow-sorted wheat chromosomes. *Nucleic Acids Res* **20**: 1897-1901.

Weschke W, Bassüner R, Hai NV, Czihal A, Bäumlein H, Wobus U (1988) The structure of a *Vicia faba* vicilin gene. *Biochem Physiol Pflanzen* **183**: 233-242.